

SOURAV KUMAR

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PROFESSIONAL SUMMARY

Machine Learning Engineer (fresher) with hands-on experience building production-grade AI systems. Developed Enterprise-FraudShield-AI—a real-time fraud detection platform processing 10,000+ transactions/sec with 94% accuracy—and built LLM-powered applications using RAG pipelines and fine-tuned models. Proficient in Python, FastAPI, TensorFlow, Apache Kafka, and AWS. IBM-certified in Machine Learning; credentials from Harvard CS50AI and Microsoft Azure. Seeking ML Engineer roles to build scalable, real-world AI solutions.

TECHNICAL SKILLS

Languages: Python, Java, Kotlin, JavaScript, SQL, C, C++

AI / ML / NLP: LangChain, RAG, LLMs, Hugging Face, Fine-tuning, Prompt Engineering, YOLO, ResNet, MobileNetV3

ML Frameworks: Scikit-learn, XGBoost, TensorFlow, LSTM, ARIMA, Prophet, Random Forest

Data & Analysis: Pandas, NumPy, Matplotlib, Seaborn, EDA, Feature Engineering, Hyperparameter Tuning

Backend & APIs: FastAPI, Flask, Django, Node.js, REST APIs, Microservices, Async Processing

Cloud & DevOps: AWS (S3, EC2, Lambda), Docker, Jenkins, Apache Kafka, Redis, Git

Model Evaluation: MAE, MSE, RMSE, R^2 Score, Confusion Matrix, ROC-AUC, Precision, Recall

PROFESSIONAL EXPERIENCE

Machine Learning Intern

IBM | Remote | June 2025 – July 2025

- Preprocessed and cleaned 10,000+ healthcare records, reducing data inconsistencies by 30% and improving model accuracy by 8%.
- Performed EDA identifying top cost-driving factors (BMI, age, smoking status), informing feature selection and model architecture.
- Developed and compared regression models (Linear Regression, Decision Tree, Random Forest), achieving 87% accuracy for medical insurance charge prediction.
- Built end-to-end predictive pipeline from raw data ingestion to deployment using Python, Pandas, and Scikit-learn.
- Reduced prediction error from baseline by 15% through hyperparameter tuning and cross-validation strategies.

Data Analysis Volunteer

Samadhaan Foundation | On-site | June 2023 – August 2023

- Analyzed program effectiveness data from 200+ community beneficiaries, identifying engagement patterns.
- Built Excel dashboards and visualizations tracking outreach metrics, improving organizational efficiency by 25%.
- Supported digital management initiatives, automating manual data entry processes.

PROJECTS

Enterprise-FraudShield-AI

Python, FastAPI, Apache Kafka, Redis, XGBoost, Docker | December 2025

- Engineered real-time fraud detection system processing 10,000+ transactions/sec, reducing detection latency by 60% vs baseline.
- Implemented ML models (XGBoost, anomaly detection, behavioral analytics) achieving 94% fraud accuracy on financial transaction datasets.
- Built production-ready microservices with Docker and Uvicorn, enabling scalable deployment and monitoring with async REST APIs.
- Integrated behavioral analytics pipeline to detect refund abuse and suspicious patterns across distributed enterprise data streams.
- Estimated business impact: prevented \$2-5M in annual fraud losses based on fraud detection rates and transaction volume.

Stock Price Prediction Models

Python, LSTM, ARIMA, Prophet, TensorFlow, Matplotlib | August 2025 – September 2025

- Developed and benchmarked LSTM, ARIMA, and Prophet models for stock forecasting, improving prediction accuracy by 22% through ensemble methods.
- Performed hyperparameter tuning and cross-validation across 5 stock datasets, reducing overfitting and increasing generalization.
- Built interactive EDA dashboards visualizing historical trends, volatility patterns, and model performance metrics.
- Evaluated models using RMSE, MAE, and directional accuracy metrics.

Fish Species Classifier (Transfer Learning)

Python, MobileNetV3, TensorFlow, OpenCV | May 2024

- Trained image classification model using MobileNetV3-Large transfer learning on 9,000 images across 9 fish species, achieving 90% overall accuracy.
- Built data augmentation and preprocessing pipelines (80/20 train-val split) ensuring robust generalization across varied datasets.
- Reduced manual classification time by 85%, automating a previously manual inspection bottleneck.
- Evaluated model using confusion matrix and per-class precision/recall metrics.

EDUCATION

B.Tech, Computer Science and Engineering (AI/ML Specialization)

University of Petroleum and Energy Studies (UPES), Dehradun | Expected May 2026

CERTIFICATIONS & CREDENTIALS

- Harvard CS50AI: Introduction to Artificial Intelligence with Python — 2024
- IBM Certified: Healthcare Cost Prediction using Machine Learning — 2025
- Microsoft Certified: Azure AI Fundamentals (AI-900) — 2026
- Data Structures and Algorithms — Coursera — 2025
- Python Programming — CodeWithHarry — 2026

ACHIEVEMENTS & ACTIVITIES

- CSI Hackathon — 2x participant and prize winner; delivered innovative team-based AI product solutions.
- Competitive Programmer — Active on LeetCode and CodeChef; consistently solving algorithmic problems under timed conditions.
- Coding Club UPES (2022–2026) — Contributing member; organized technical events and represented club in competitions.
- TechFest 2023, UPES — Volunteered in event coordination and technical management.